

Scene-informed optimization of measurement locations for radiological assessment

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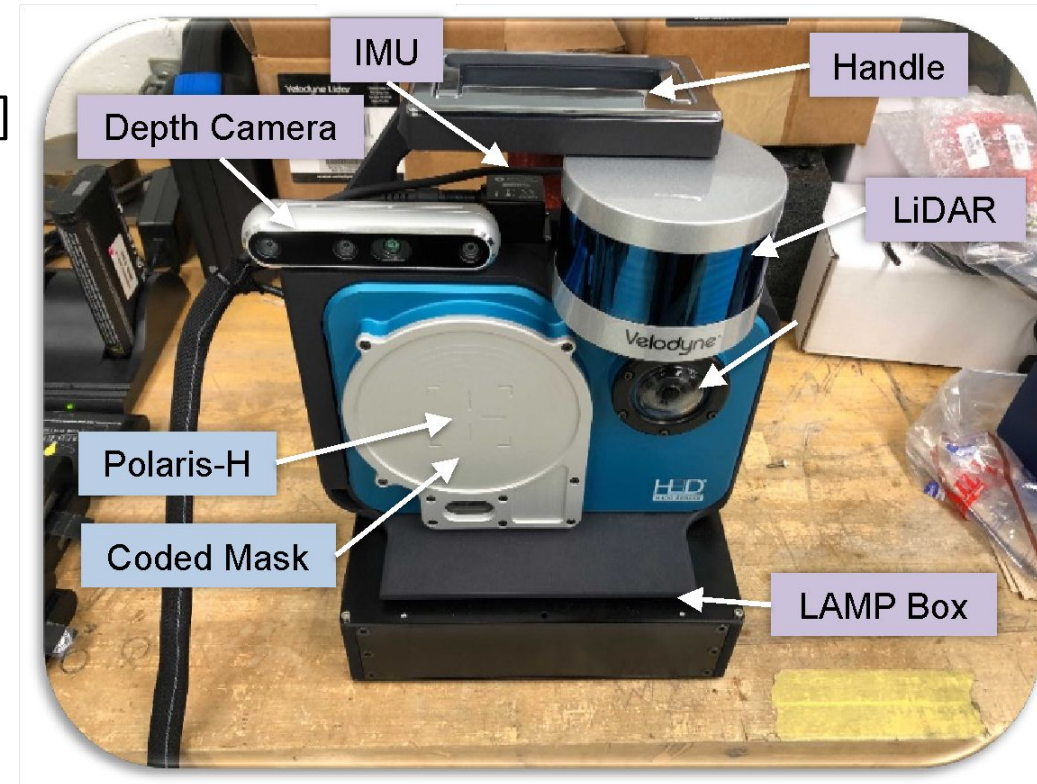
Motivating Application: Safeguards

- Develop a technology that can validate amounts and types of different nuclear materials present in monitored items for safeguards
- Recently, IAEA has passed gamma-ray imaging on for more extensive evaluation –application of 3D mapping combined with radiation imaging
- Passive gamma ray imaging is a *non-destructive* method can provide a quantitative measure of the nuclear materials present
- Safeguards analysts have a limited amount of time to perform measurements, ideally with minimal disruption to facility operation



H420-LAMP and Scene Data Fusion

- Commercially available H420 detector fitted with LBNL's Localization and Mapping Platform (LAMP)
 - H420: commercial gamma-ray imager developed by H3D [1]
 - LAMP: contextual sensor package for free-moving 3D reconstruction [2]
 - Simultaneous Localization and Mapping (SLAM) with Google Cartographer [3]
- Portable radiation detection systems can be equipped with contextual sensors to allow free-moving 3D gamma-ray imaging through a method called **Scene Data Fusion (SDF)**



H420-LAMP Imaging System

[1] C. G. Wahl *et al.*, "The Polaris-H Imaging Spectrometer," *Nucl. Instrum. Methods A*, vol. 784, pp. 377–381, Jun. 2015.

[2] R. Pavlovsky *et al.*, "3-D Radiation Mapping in Real-Time with the Localization and Mapping Platform LAMP from Unmanned Aerial Systems and Man-Portable Configurations," *arXiv:1901.05038*, 2018.

[3] W. Hess, *et al.*, "Real-Time Loop Closure in 2D LIDAR SLAM," in *Proc. IEEE Int. Conf. Robot. Auto.*, pp. 1271–1278, 2016.

Measurement Procedure of H420 LAMP

Build 3D model of scene

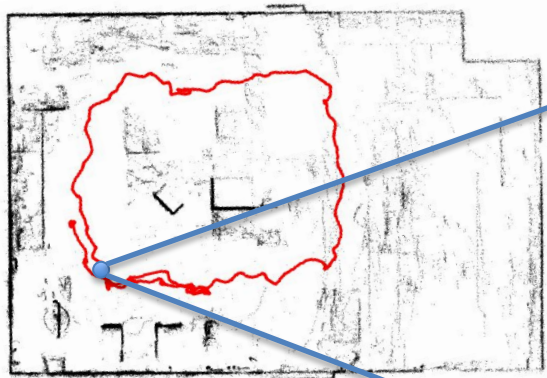
- ~1 minute walkaround
- Establish object of interest
- Identify static measurement positions

Static Measurements

- Point and dwell
 - High statistics
 - Reduce pose uncertainty
- Repeat as prescribed

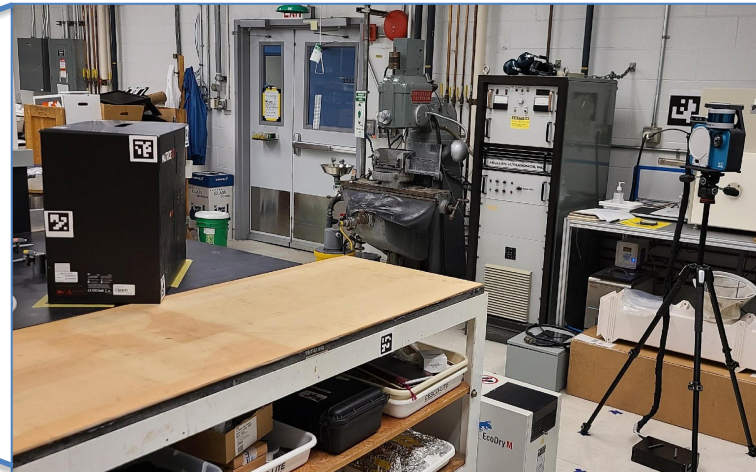
Global Analysis

- Quantitative 2D imaging
 - Compton and Coded aperture
 - Use depth information
- 2D to 3D analysis
- Estimate material quantity



xy view of 3D LiDAR point cloud

H420-LAMP in Static Measurement Position



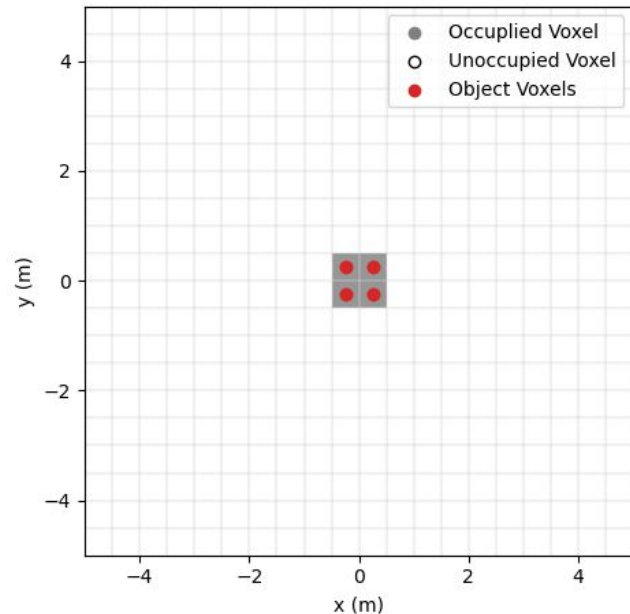
Optimization Approach

- 1. Determine object of interest**
 - Manually or through 3D object detection
- 2. Find all plausible measurement positions**
- 3. Calculate sensitivity to object voxels at each measurement position**
- 4. Find subset of positions that maximize metric of interest**

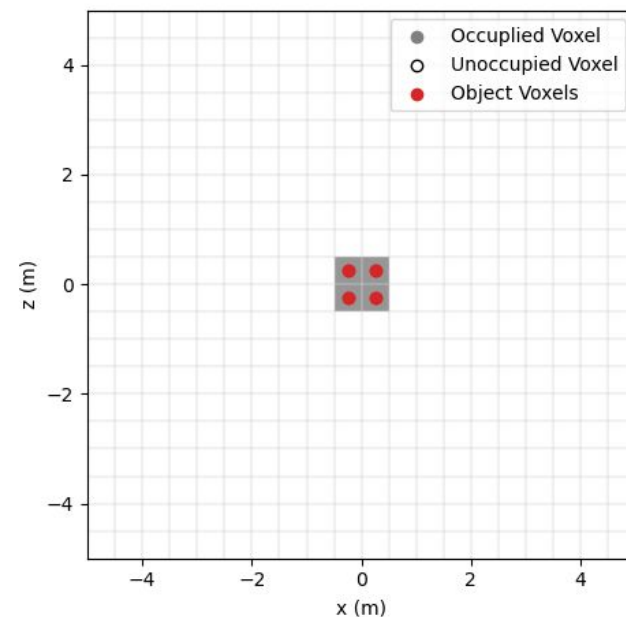
Determine Object of Interest

- Currently using **manual** object identification
- 10 m x 10 m x 10 m voxel grid - 8,000 voxels with a 0.5 m pitch
- 1 m x 1 m x 1 m object of interest in center of vg
 - 8 occupied source voxels, all other voxels are unoccupied

xy plane

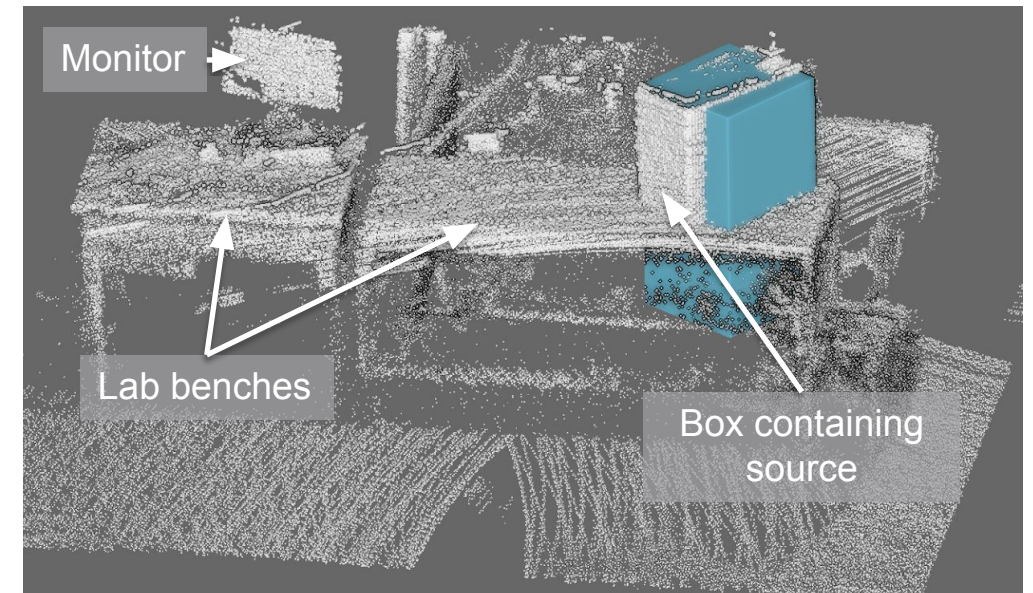


xz plane



- Planning to use **automatic** object identification with 3D object detection for field measurements
- Current results are promising but need refinement

Votenet model pre-trained on sunrgbd data [1] using MMDetection3D open-source object detection toolbox [2]

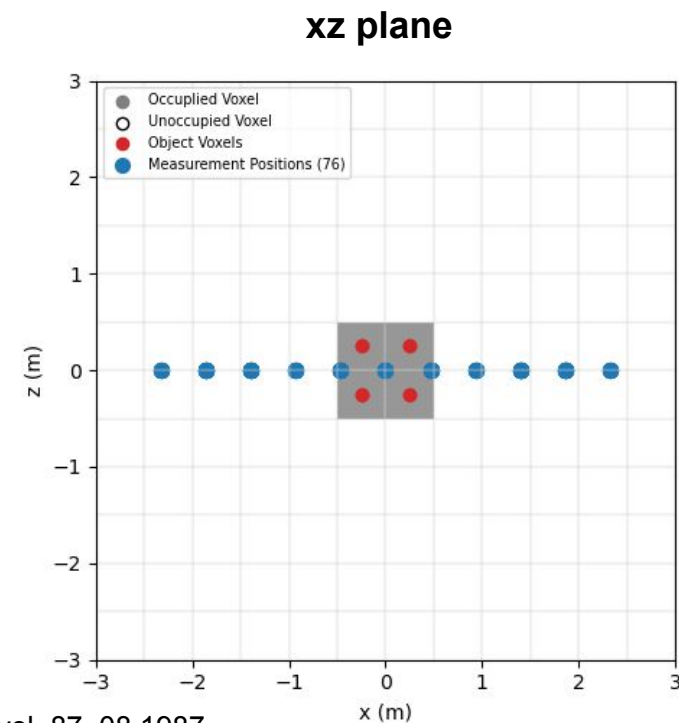
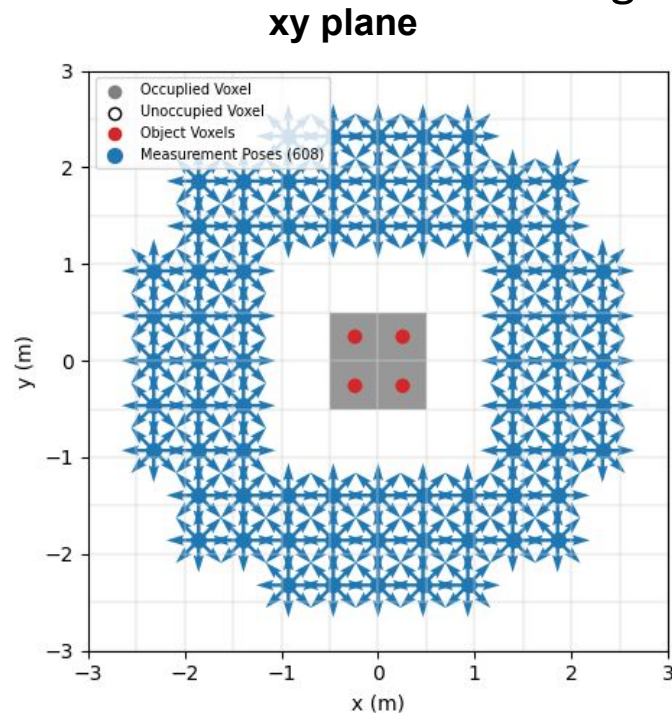


[1] C. Qi *et al.*, (2019). "Deep Hough Voting for 3D Object Detection in Point Clouds". Proceedings of the IEEE International Conference on Computer Vision.

[2] MMDetection3D Contributors, (2020). "MMDetection3D: OpenMMLab Next-Generation Platform for General 3D Object Detection". <https://github.com/open-mmlab/mmdetection3d>.

Find all possible measurement positions

- 76 potential measurement positions found in xy plane
 - $Z = 0$ for all positions, 8 potential orientations around Z-axis, $76 \times 8 = \mathbf{608 \text{ unique poses}}$
- Measurement location must be within desired standoff distance, be physically feasible, and have a direct line of sight to at least one of the object voxels
 - Direct line of sight established through ray-tracing [1]

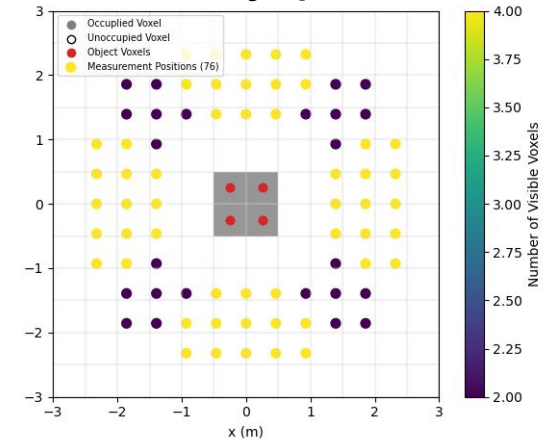


[1] J. Amanatides and A. Woo, "A fast voxel traversal algorithm for ray tracing," Proceedings of EuroGraphics, vol. 87, 08 1987

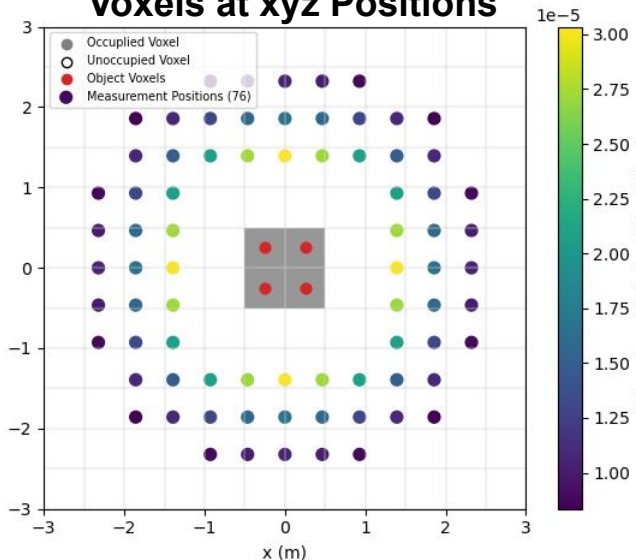
Calculate Sensitivity

- Calculate sensitivity to each object voxel at each potential measurement location
 - Sensitivity = response * $1/r^2$ * attenuation
 - Assume 662 keV doubles for response
 - Use ray tracing algorithm [1] to inform attenuation estimates - dummy attenuation value for occupied source voxels (0.9 m^{-1})

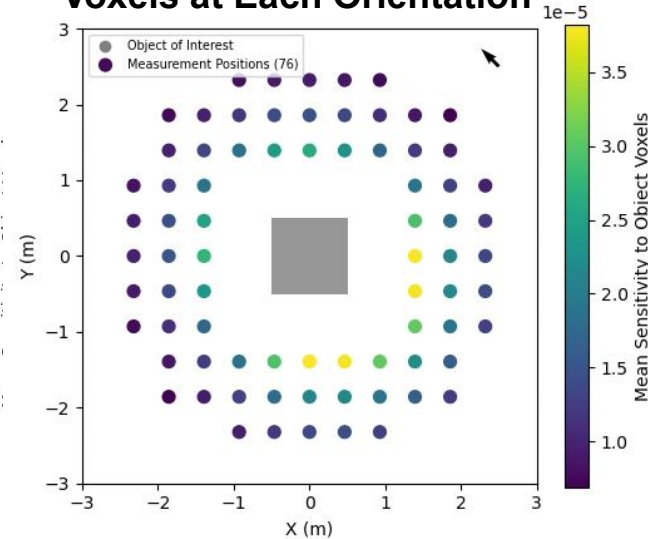
Number of Visible Voxels at each xyz position



Mean Sensitivity to Object Voxels at xyz Positions



Mean Sensitivity to Object Voxels at Each Orientation



Semantic segmentation with Google Deeplab ADE20K-trained model [2]



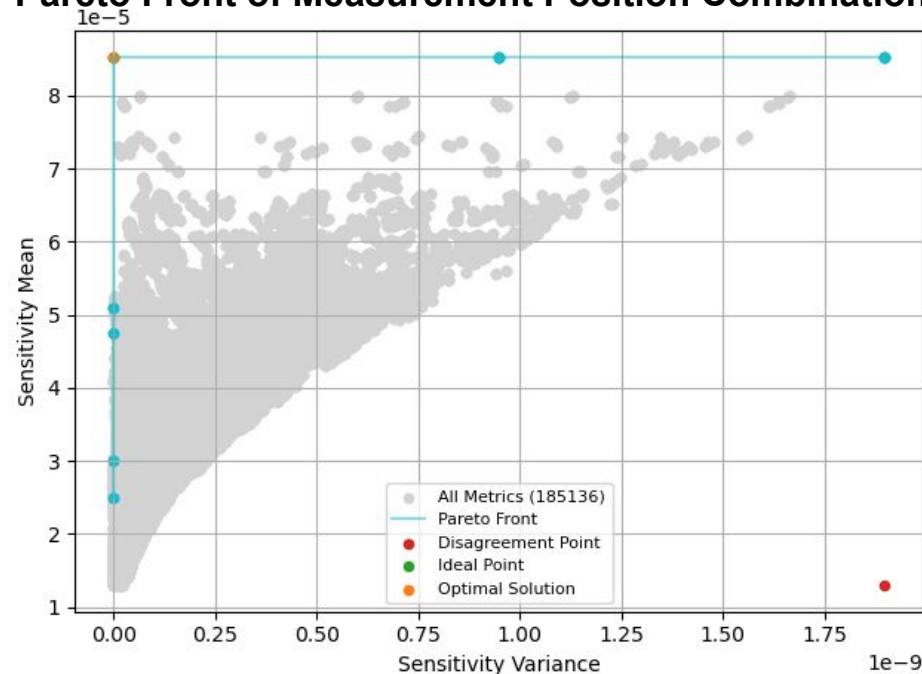
[1] J. Amanatides and A. Woo, "A fast voxel traversal algorithm for ray tracing," Proceedings of EuroGraphics, vol. 87, 08 1987

[2] L.-C. Chen et. al. (2018), "Encoder-decoder with atrous separable convolution for semantic image segmentation," in Proc. Eur. Conf. Comput. Vis. (ECCV), pp. 801–818

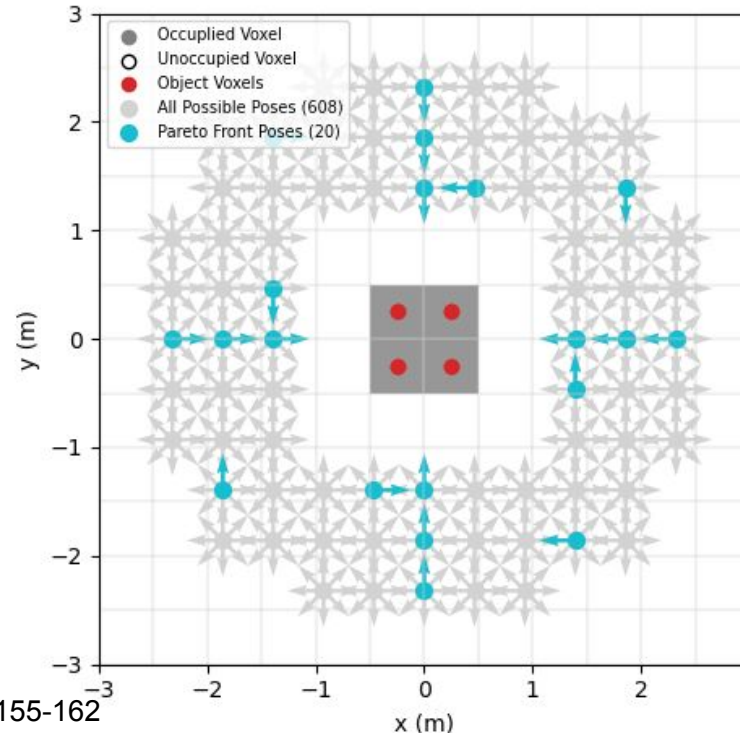
Find Subset of Positions that Maximize Objective Function

- Multi objective optimization – minimize sensitivity variance and maximize sensitivity mean
 - Find Pareto front of all metrics – all solutions on the Pareto front may be considered optimal
 - Nash Bargaining Solution can be used to select solution from the Pareto front [1]
- For the best combination of positions, the optimization must be performed on all possible combinations of poses – for 640 poses and combinations of 2 positions, this results in 205,120 possible combinations

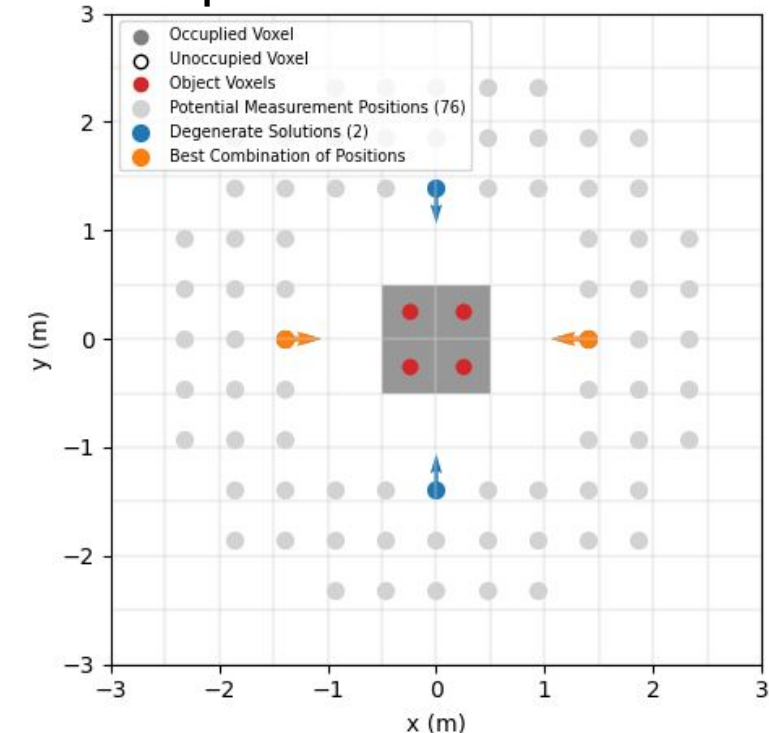
Pareto Front of Measurement Position Combinations



Pareto Front Poses



Optimal Combination of Poses

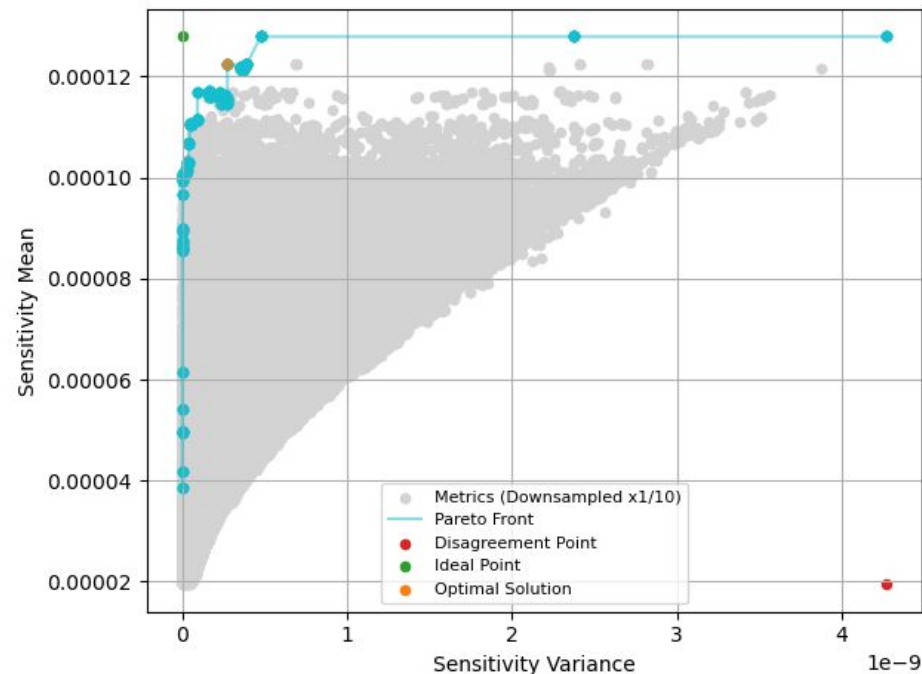


[1] J. Nash, "The Bargaining Problem", Econometrica, vol. 18, no. 2, p.155-162

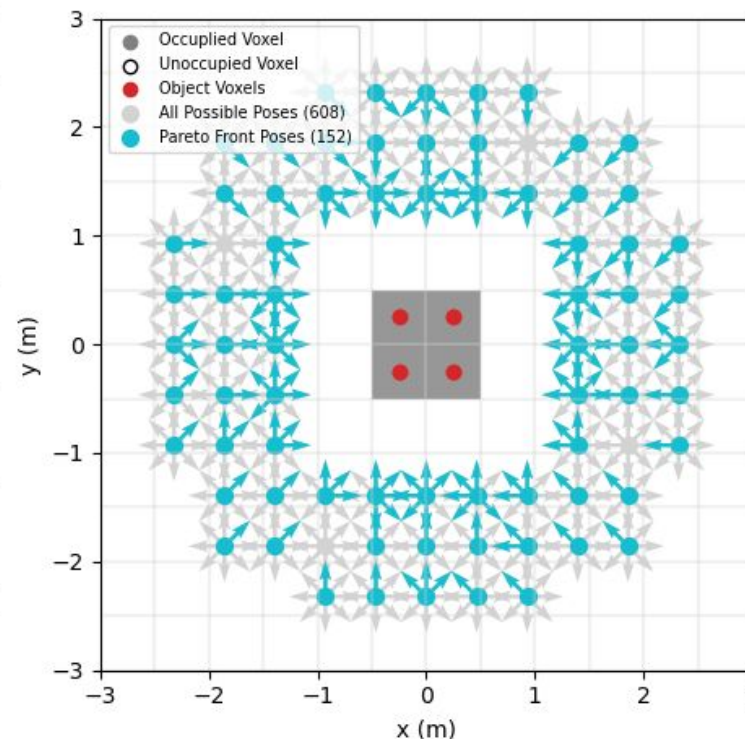
Find Subset of Positions that Maximize Objective Function

- For 640 poses and combinations of 3 positions, this results in **43,895,680** possible combinations
 - Computation time grows very long, even for this very simple measurement scenario
- Pareto front shown for $P = 3$

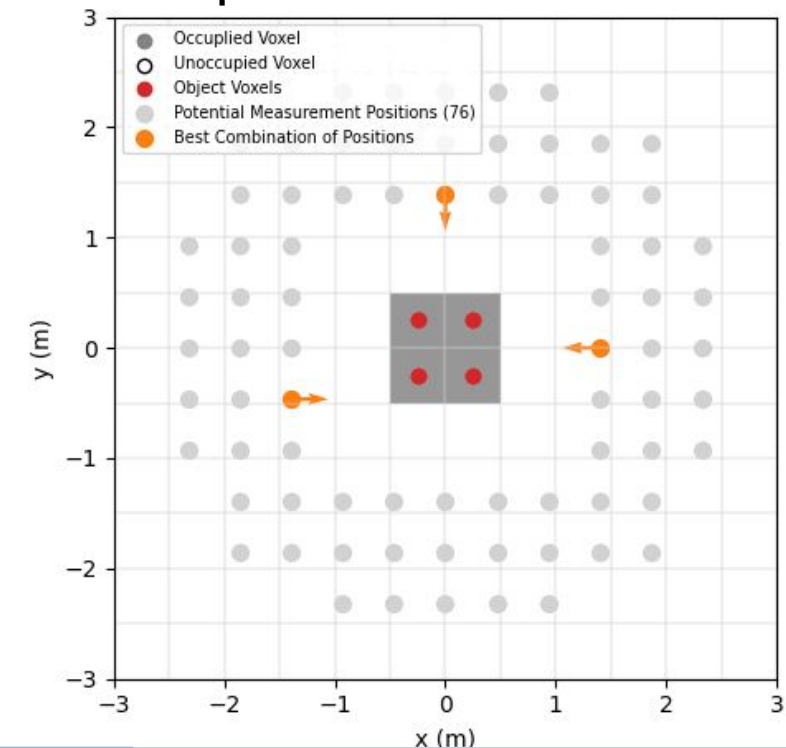
Pareto Front of Measurement Position Combinations



Pareto Front Poses

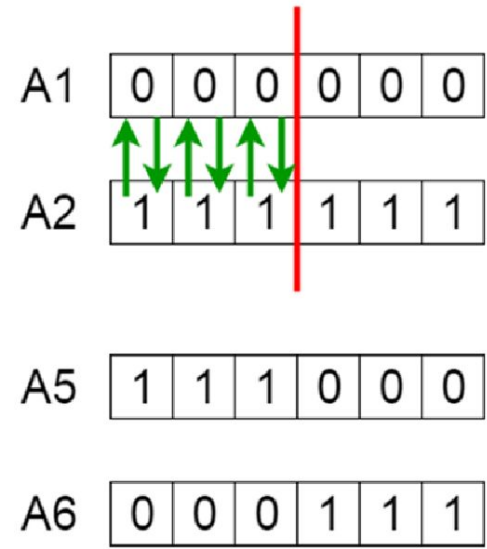
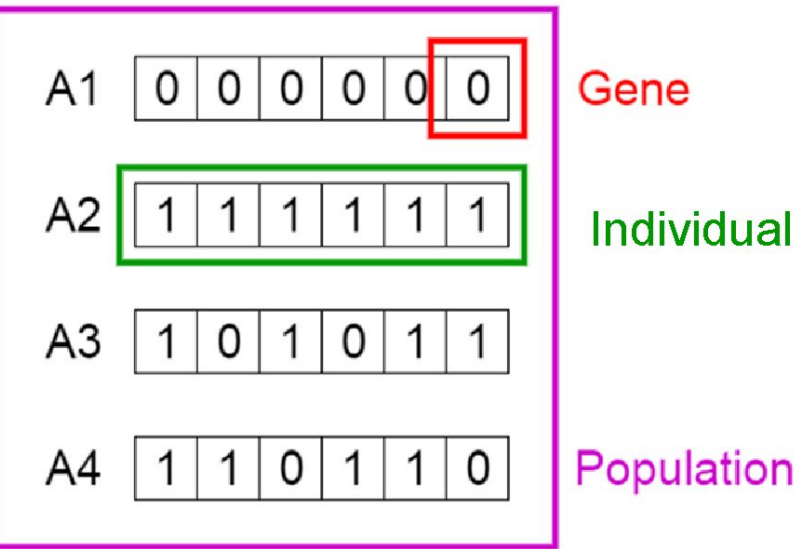


Optimal Combination of Poses



Optimization Frameworks: Genetic Algorithm (GA)

1. Initial Population
2. Fitness function
3. Selection
4. Crossover
5. Mutation
6. Termination



Gene	Individual	Population
656*	1222	1501
261	1155	1475
314	978	1378
281	627	654
43	197	1143
1008	1021	1241
57	625	1182
15	624	1268
53	247	706
65	1072	1331
3	598	478
800	65	1
55	32	432
70	275	641
901	84	7

*Indices for measurement positions

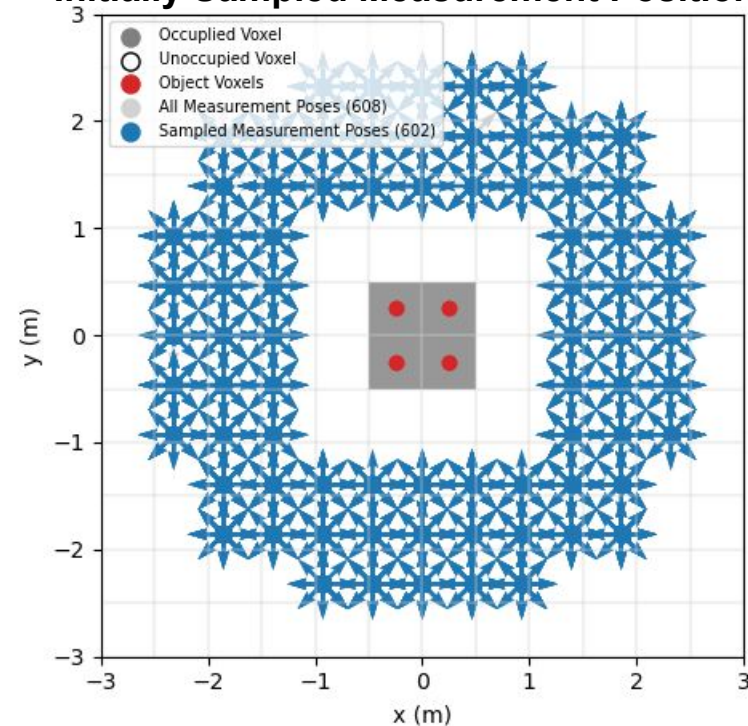
Diagram of Genetic Algorithm [1]

[1] V. Mallawaarachchi "Introduction to Genetic Algorithms — Including Example Code", <https://towardsdatascience.com/introduction-to-genetic-algorithms-including-example-code-e396e98d8bf3>

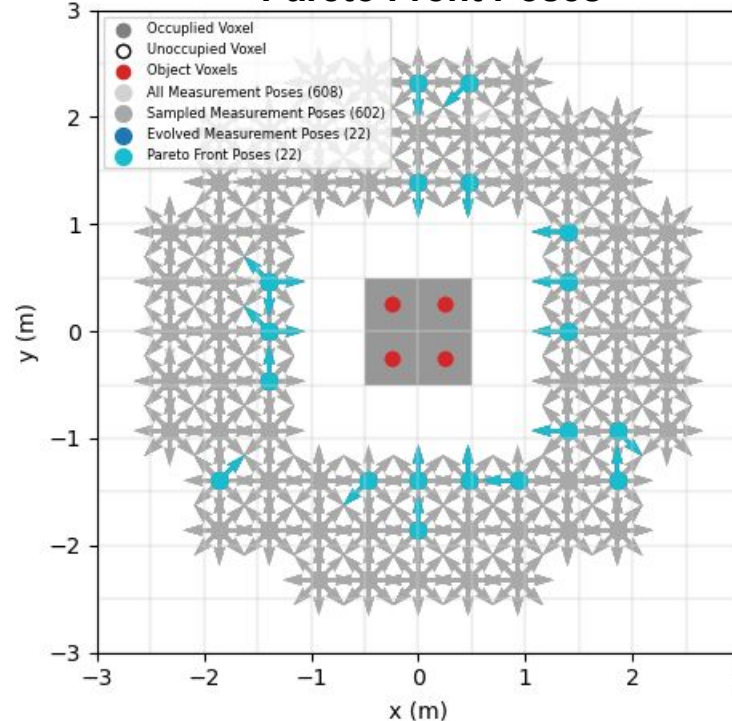
Use Genetic Algorithm to Evolve

- Using Non dominated Sorting Genetic Algorithm (NSGA-II) from pygmo [1]
 - 50 generations, 0.95 crossover probability, 0.2 mutation probability
- For $N=3$, initial population of 1000 combinations of 3 poses (43,895,680 total possible combinations)
- 602/608 poses represented in initial sample

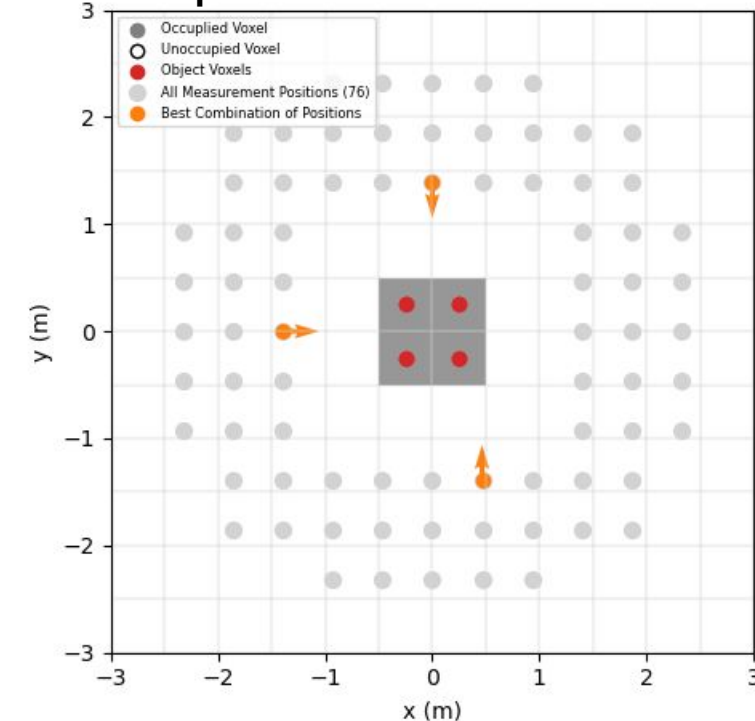
Initially Sampled Measurement Positions



Pareto Front Poses



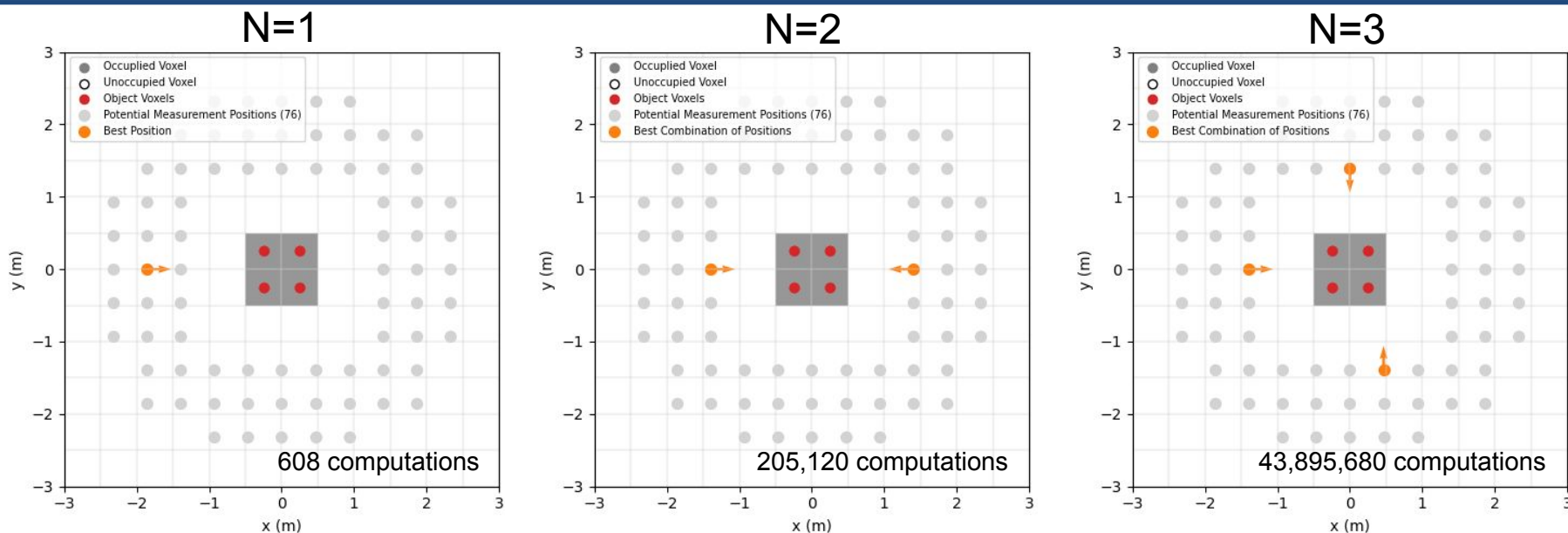
Optimal Combination of Poses



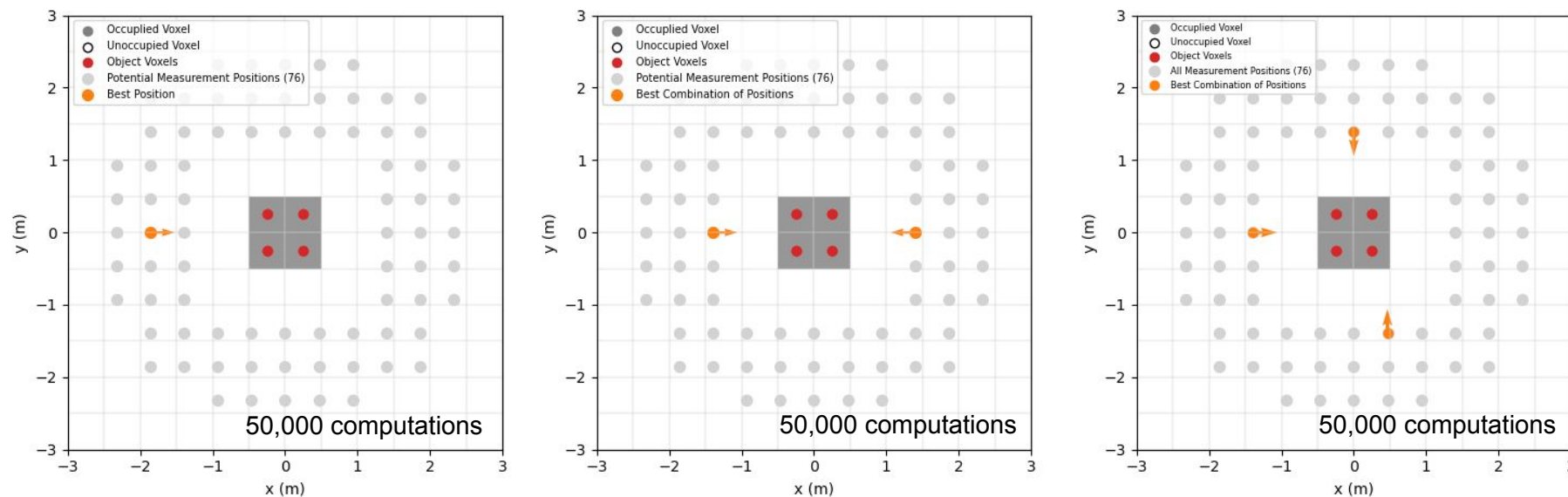
[1] F. Biscani and D. Izzo "A parallel global multiobjective framework for optimization: pagmo", Journal of Open Source Software, vol.5 no. 53, p 2338.

Brute Force vs GA

Brute Force



GA



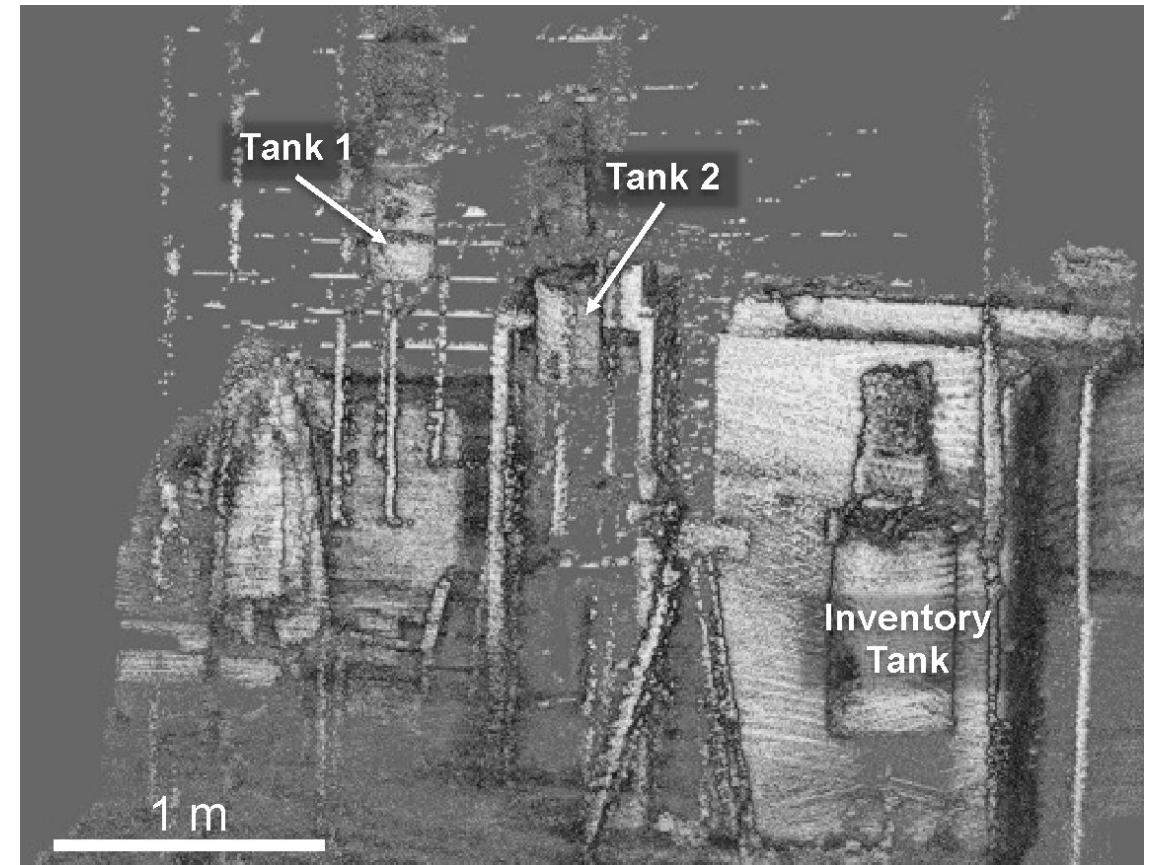
Field measurement

- Mock material holdup loop
- 0.5 mCi F-18 and 0.5 mCi of Tc-99m diluted in 10 gallons (37.9 L) of water

Picture of mock holdup loop

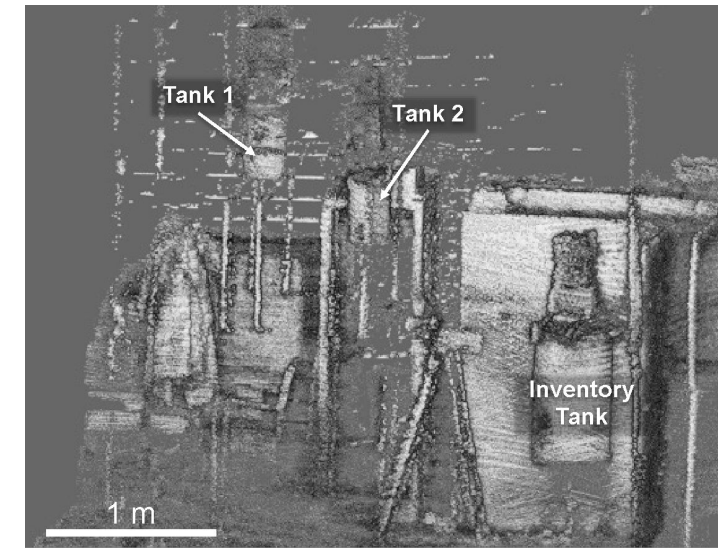


LiDAR point cloud of mock holdup loop

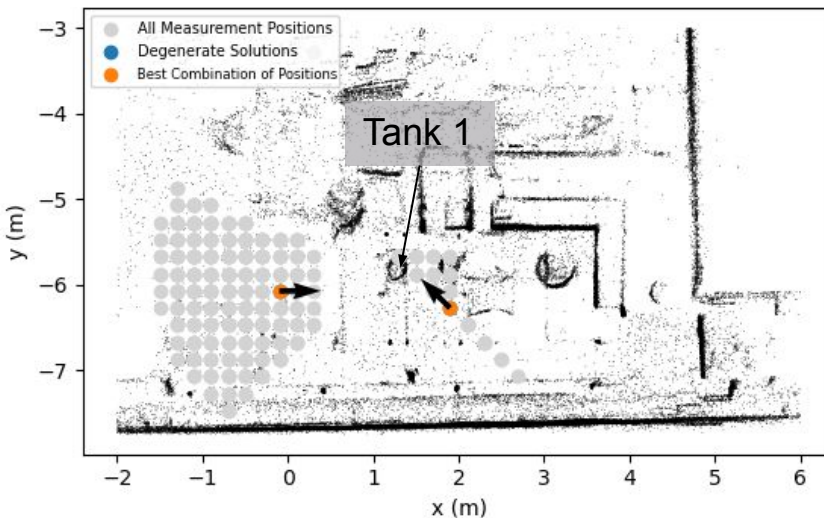


Field measurement

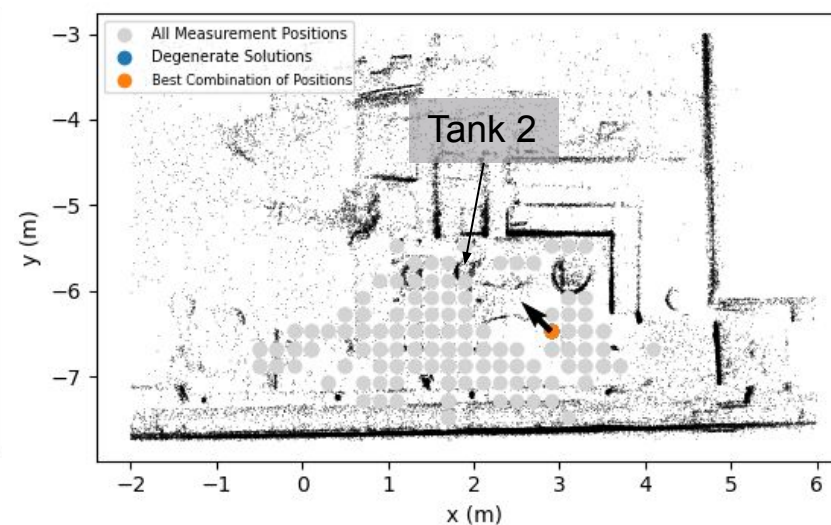
- Optimization performed for each of the three tanks on 511 keV line of F-18
- Many objects present in the scene, making possible measurement positions quite limited



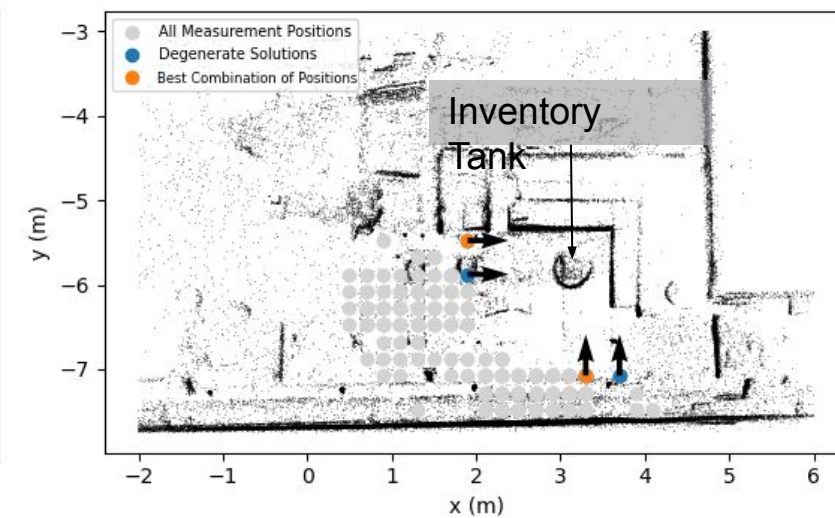
Tank 1 Optimal Positions



Tank 2 Optimal Positions



Inventory Tank Optimal Positions

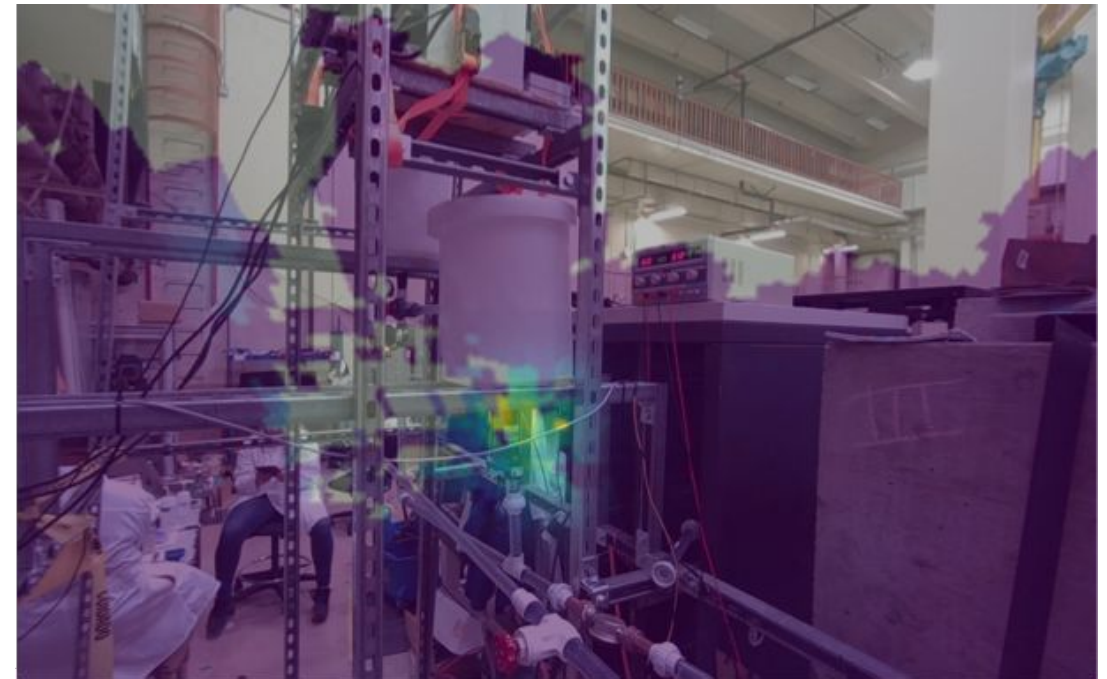


Field measurement images of Tank 2

- 8.7 L (2.3 gallons) of water in tank 2, remaining water in inventory tank
- Imager was placed within tolerance (± 0.1 m, $\pm 22.5^\circ$) at optimal measurement position and measurement taken for 20 minutes
- Preliminary image could use refinement – co-registration, use of GPP algorithm [1], include attenuation



Image of Tank 2 from Depth Camera During Measurement



Preliminary 511 keV Relative Activity Reconstruction

[1] J. Lee N-11-05 and N-11-06 at IEEE NSS 2022

Conclusions and future work

- Accomplishments so far
 - Brute Force and Genetic Algorithm arrive at the same solutions, with GA being much faster
 - Optimization chooses sensible measurement positions in real scenarios
 - Field measurement scenarios allowed localization of source activity to relevant tanks
- Remaining goals
 - Real time localization mode to ensure imager is placed in 'optimal' position within tolerance
 - Machine learning techniques will be used to improve attenuation estimates and object position identification
 - Quantification of source activity from material containers

Thank you!

Questions?

This material is based upon work supported by the Department of Energy National Nuclear Security Administration through the Nuclear Science and Security Consortium under Award Number DE-NA0003996.

Background image: LiDAR point cloud of Pripjat Town Center from K. Knecht *et al.*, “3D Compton Imaging of Distributed Sources around the Chernobyl Nuclear Power Plant”, 2021 IEEE NSS/MIC.